SEFI Canonical Mathematics (Physics Core)

Jason Dutton
sefidefi@outlook.com

August 2026

Contents

1 SEFI Canon Overview **1**

2 Identity Continuity **1**

3 SEFI Fields **2**

4 Warp-Metric Scalar **2**

5 SEFI Lagrangian **2**

6 Field Equations **2**

7 Warp-Metric Dynamics **3**

8 FTL-Compatible Warp Solutions **3**

9 Projection Law **3**

10 Quantization **3**

11 Prediction Skeleton **3**

1 SEFI Canon Overview

SEFI is a geometric physics framework built on identity continuity, tension fields, torsion drift, coherence, curvature-knot dynamics, and warp-metric geometry.

2 Identity Continuity

$$I[x] = \int \sqrt{g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}}\, d\tau$$

SEFI postulate:

$$\delta I = 0$$

under SEFI tension-projection and warp-metric transformations.

3 SEFI Fields

$$T(x), \quad \Theta_{\mu\nu}(x), \quad \Gamma_\mu(x), \quad K(x), \quad g_{\mu\nu}(x)$$

4 Warp-Metric Scalar

$$W = R + \alpha T^2 + \beta \Theta_{\mu\nu}\Theta^{\mu\nu} + \gamma \Gamma_\mu \Gamma^\mu - \delta K^2$$

5 SEFI Lagrangian

$$\mathcal{L} = \frac{1}{2\kappa} W + \frac{1}{2}(\nabla T)^2 + \frac{1}{4}(\nabla \Theta)^2 + \frac{1}{2}(\nabla \Gamma)^2 - V$$

$$V = \lambda_T T^4 + \lambda_\Theta (\Theta^2)^2 + \lambda_\Gamma (\Gamma^2)^2 + \lambda_K (K^2 - K_c^2)^2$$

6 Field Equations

Metric:

$$\frac{1}{2\kappa} G_{\mu\nu} = T_{\mu\nu}^{SEFI}$$

Tension:

$$\Box T + \alpha T + \frac{\partial V}{\partial T} = 0$$

Torsion:

$$\Box \Theta_{\mu\nu} + \beta \Theta_{\mu\nu} + \frac{\partial V}{\partial \Theta^{\mu\nu}} = 0$$

Coherence:

$$\Box \Gamma_\mu + \gamma \Gamma_\mu + \frac{\partial V}{\partial \Gamma^\mu} = 0$$

Curvature-knot:

$$-2\delta K + \frac{\partial V}{\partial K} = 0$$

7 Warp-Metric Dynamics

$$W = \text{const}$$

Stability:

$$\frac{\delta^2 S}{\delta \phi^2} > 0$$

8 FTL-Compatible Warp Solutions

$$I[x] = \text{invariant}, \quad g_{\mu\nu} \to \tilde{g}_{\mu\nu}$$

Causality preserved globally.

9 Projection Law

$$F_i = P_i(T, \Theta, \Gamma, K)$$

10 Quantization

Promote fields to operators. Canonical commutators. Hamiltonian:

$$\mathcal{H} = \Pi_\phi \dot{\phi} - \mathcal{L}$$

Identity continuity constraint:

$$\hat{I}|\Psi\rangle = I_0|\Psi\rangle$$

11 Prediction Skeleton

$$\mathcal{O}_k[g, T, \Theta, \Gamma, K]$$

Cosmology, particle physics, gravitational waves, rotation curves.